

Global Cybersecurity Threats – Visual Analytics & Insights

1. Project Overview

The objective of this project is to analyze **global cybersecurity threats** using **Tableau** and apply **visual analytics principles** to uncover patterns, trends, and risk factors across regions, industries, attack types, and time.

This project focuses on transforming raw cybersecurity data into **meaningful, interactive visualizations** that support:

- Threat awareness
 - Risk assessment
 - Decision-making for cybersecurity teams
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2. Dataset Description

The dataset contains records of cybersecurity incidents across different countries and industries. Key attributes include:

- **Country / Region**
- **Industry**
- **Year / Time**
- **Attack Type** (e.g., Malware, Phishing, DDoS)
- **Impact Level**
- **Financial Loss**
- **Number of Affected Users**
- **Severity / Risk Score**

The dataset was imported directly into **Tableau Desktop (.twbx)**, allowing packaged data and dashboards to be shared seamlessly.

3. Data Discovery (Exploratory Analysis)

Before answering business or security questions, visual exploration was performed to understand the data structure and relationships.

Exploratory Visualizations Used:

- Bar charts to compare attack frequency
- Line charts to observe yearly trends
- Heatmaps to identify high-risk regions and industries
- Geographic maps to visualize global threat distribution

These visualizations helped identify:

- Regions with consistently high cyber incidents
 - Growth trends in cyberattacks over time
 - Industries most affected by specific attack types
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4. Key Questions Answered Using Visualizations

Q1. Which regions experience the highest number of cyberattacks?

Using a **map visualization**, it was observed that certain regions consistently experience higher volumes of cybersecurity incidents, indicating higher exposure or reporting maturity.

Q2. How have cybersecurity threats changed over time?

A **time-series line chart** shows a steady increase in cyber incidents, reflecting the growing digital footprint and sophistication of attacks.

Q3. Which industries are most targeted?

A **bar chart combined with color encoding** highlights industries such as finance, healthcare, and IT as frequent targets due to sensitive data and financial incentives.

Q4. What types of cyberattacks are most common?

A **stacked bar chart** compares different attack types across industries, revealing that phishing and malware dominate most sectors.

Q5. Which attack types result in the highest impact or loss?

Using **size and color encoding**, high-impact attacks were identified even if their frequency was low, emphasizing that impact matters more than volume alone.

5. Calculations Used in Tableau

The project uses multiple types of Tableau calculations:

1. Aggregate Calculations

Used for:

- Total attacks per region
- Sum of financial loss
- Average severity score

Example:

```
SUM([Financial Loss])
```

2. Table Calculations

Used to analyze:

- Year-over-year growth
- Percentage contribution of attack types

Example:

```
(SUM([Attack Count]) / TOTAL(SUM([Attack Count]))) * 100
```

3. Calculated Fields

Custom measures such as:

- Risk level classification
- Impact categories (Low / Medium / High)

These calculations help convert raw metrics into **decision-oriented insights**.

6. Visual Analytics Principles Applied

- **Color Encoding** to represent severity and impact
- **Size Encoding** to show magnitude of losses
- **Consistency** across dashboards for better cognitive processing
- **Minimal clutter** to keep insights clear
- **User-driven filtering** for interactivity

Each visualization was chosen deliberately to match the data and question being asked.

7. Dashboards & Interactivity

The final solution consists of **interactive dashboards** allowing users to:

- Filter by region, industry, year, or attack type
- Drill down into specific countries or sectors
- Compare trends side-by-side

This allows stakeholders to explore *what-if scenarios* without technical complexity.

8. Insights & Findings

Key Findings:

- Cybersecurity incidents are increasing globally over time
 - Certain industries are consistently more vulnerable
 - High-impact attacks may be less frequent but cause disproportionate damage
 - Regional differences suggest uneven cybersecurity preparedness
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9. Limitations

- Dataset depends on reported incidents; underreporting may exist
 - Severity scoring may be subjective
 - Lack of real-time data limits predictive analysis
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10. Conclusion

This Tableau project demonstrates how **visual analytics** can transform complex cybersecurity data into actionable insights. By combining effective visualization techniques, calculations, and interactivity, the project supports better understanding of global cyber risk patterns and helps inform strategic cybersecurity planning.